



Deep structural enhanced network for document clustering

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Abstract

Recently, deep document clustering, which employs deep neural networks to learn semantic document representation for clustering purpose, has attracted increasing research interests. Traditional deep document clustering models rely only the document internal content features for learning the representation and suffer from the insufficient problem of representation learning. In this paper, we introduce a deep structural enhanced network for document clustering, namely DSEDC. The DSEDC model enhances the AE-based internal document representation with GCN-based external structural document semantics for achieving better clustering performance. An ensemble-reinforced enhancement strategy is designed, in which a complete document representation, captured by fusing document internal semantics and external semantics, and an enhanced document internal representation, captured with the help of complete document representation, are learned in a layer-by-layer reinforcement manner. Extensive experiments demonstrated that our proposed DSEDC model performs substantially better than state-of-the-art deep document clustering models.

Keywords Document clustering · Deep clustering · Graph convolutional network · Semantic representation

1 Introduction

With the exponential growth of information on the World Wide Web and the wide availability of documents,

document clustering, which aims to discover the underlying structure of document datasets, acts as a fundamental tool and is desperately needed for many applications. Recently, deep document clustering has attracted significant attention and achieved state-of-the-art performance. One key idea of most deep document clustering models is relying on deep neural networks to map each data sample to a low-dimensional hidden data representation space, normally topic space, for discovering data partitions. Therefore, learning effective data representation is important for deep document clustering.

Traditional deep document clustering models only make use of the content features of each data sample to learn the semantic data representation [1–4]. As a result, document samples are represented internally with the consideration of those features within each document sample. However, because each document has a limited length and it tends to flexibly express its meaning in various ways, only making use of internal content features leads to incomplete document representation learned. Each document is described by a subset of discriminative features, while other crucial discriminative features are missing. As a result, similar documents are frequently described by different sets of discriminative features and regarded as not close. The incomplete document representation leads to inaccurate estimation of deep

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document clustering. Internal data representations are not sufficient to obtain accurate estimation of clustering assignments.

In addition to those internal semantic context features, there is some other useful information that can be discovered externally outside each document. For example, close neighbors of documents, regarded as document external structural information, usually carry complementary meaningful but missing content features of text documents. Therefore, it is useful to make use of document external structural information to improve its internal semantic data representation. Although the external structural information is useful, there has been little work applying it to enhance the internal semantic document representation for improving document clustering performance.

In this paper, we design an end-to-end deep document clustering model, namely DSEDC, to learn document partitions with the help of a structural enhanced network. Specifically, we take the internal document semantics as the base and enhance them by using document external structural semantics. For each document, an autoencoder (AE) [5] is used to capture its internal data semantic representation by mapping original high-dimensional internal context features to a lower-dimensional representation space. We employed the graph convolutional network (GCN) [6] model to learn the external semantic representation of each document because it shows promising results on representing data as graphs and encoding each data sample with its external structural information. Both the AE and the GCN models are able to capture different levels of latent semantic information through their multiple layers of network architectures. To enhance the internal document representation with the external semantic information, a common strategy is to learn the document representations separately and directly combine those learning results with a normal fusion process [7–9], assuming that each document representation is independent of each other. However, in reality, the internal and external document representations are not independent but interrelated. A good external document representation can be used to emphasize, supplement, and disambiguate internal document features, which is helpful to improve internal document representation in return. Therefore, it would be useful to boost the document internal semantic representation with the help of external semantic information during the learning process. In the DSEDC model, we designed an ensemble-reinforced enhancement strategy for document representation learning, which learns an enhanced internal document representation as well as an ensemble document representation simultaneously layer by layer. The ensemble document representation learned by integrating the internal and external document representation is used to capture the complete semantic features of a document within document internal content and external

structure. The enhanced internal document representation is then learned taking the ensemble document representation into consideration. The enhanced internal document representation and the ensemble document representation are reinforced layer by layer so that the semantic information learned by the earlier stage can be used to further enhance the subsequent representation learning process. Furthermore, a joint objective function was then deployed to fulfill two tasks, learn the document partition and optimize the document representation learning networks of the DSEDC model simultaneously.

This paper makes three main contributions, which are summarized as follows:

- An end-to-end deep document clustering model, namely DSEDC, is designed to learn document partitions with the help of a structural enhanced network. Specifically, the DSEDC model takes the internal document semantics as the base and enhances them by using document external structural semantics.
- An ensemble-reinforced enhancement strategy is designed to improve the document semantic representation. An ensemble document representation is learned to obtain a complete document representation from different aspects. An enhanced internal semantic representation is then learned with the help of the ensemble document representation, the enhanced internal semantic representation and the ensemble representation are learned in a layer by layer reinforce manner.
- Extensive experiments on realistic datasets are conducted to compare our proposed DSEDC model with a number of state-of-the-art clustering models. Experimental results demonstrate that the DSEDC model is effective and substantially improves document clustering performance.

2 Related work

This paper is closely related to 2 branches of related works, in particular, deep clustering with internal semantic information and deep clustering with external semantic information. In the following paragraphs, we give a series of works that lead to our own.

2.1 Deep clustering with internal semantic information

Recently, there have been an increasing number of research studies about learning feature representation and clustering assignments via deep neural networks. The deep clustering network (DCN) [4] is a joint dimensional reduction and k-means clustering framework where the

dimensional reduction model is investigated based on a deep neural network. Deep embedding clustering (DEC) [1] was designed to learn a mapping from the data space to a lower-dimensional embedded space in which it iteratively optimizes a KL divergence-based clustering objective. The improved deep embedded clustering model (IDEC) [2] was designed by adding local structures on the basis of DEC, and [3] proposed a variational deep embedding method that is able to approximately learn the underlying document representation specified by the document generation process. Document clustering can then be conducted with the help of the learned document representations. Similarly, there are many other models [10–15] that were proposed to learn the semantic document representation that can be applied to the clustering task. However, all the above models involve only the internal semantic representation for learning document partitions. None of them consider the data sparseness problem of text documents, which leads to insufficient learning of document representation for obtaining good clustering assignments. Furthermore, none of them systematically explored the external document semantic information.

2.2 Deep clustering with external semantic information

To cope with the external semantic representation underlying the data, one typical traditional method is spectral clustering [16], which treats the samples as the nodes in a weighted graph and uses the graph structure of data for clustering. For example, the deep neural network for learning graph representations (DNGR) [17] is a spectral clustering method that uses the adjacency matrix as the similarity matrix.

In recent years, benefiting from the development of deep learning, the task of graph clustering has progressed significantly, aiming to find sets of related vertices in graphs. In most of these models, data samples are represented by their nearby graph structures modeled by deep neural networks [17, 18]. Inspired by the success of GCN, a number of recent research works [19–26] have employed the GCN model to model the data structural representation for graph clustering. For instance, the graph autoencoder (GAE) model and the variational graph autoencoder (VGAE) [19] models learn the data semantic representation with a two-layer GCN and reconstruct the adjacency matrix for each node with an autoencoder and variational autoencoder, respectively. The adversarially regularized graph autoencoder (ARGA) and the adversarially regularized variational graph autoencoder (ARVGA) [20], as variants of GAE and VGAE, were designed by incorporating an adversarial method into the GAE and VGAE models. Most of the existing graph

clustering models rely on external structural information but neglect the usefulness of internal semantic information of data samples learned by deep clustering to discover data partitions. There has been little work making the use of the above semantic information for the clustering process.

Recently, there are some research works that consider both the internal and external semantic information for aiding the clustering process [25, 26]. The structural deep clustering network (SDCN) [25] integrates the structural information into deep clustering. A delivery operator was designed to transfer the representations learned by autoencoder to the corresponding GCN layer. Based on the SDCN, the attention-driven graph clustering network (AGCN) [26] was introduced to dynamically fuse the internal and external representations by adaptively aggregating multi-scale features embedded at different layers. Despite the success of the above models, both the SDCN and AGCN model are not well suited for the document clustering task. The reason is that the two models were built to explore the usage of GCN-based structural information for the clustering task. As a result, the GCN-based structural representation, which contains a large number of noise features for text documents, is used as the base representation. Content semantics, whose importance is well recognized by previous literature, is not fully investigated but only serves as supplement information in SDCN and AGCN models. As a result, existing models are more suitable for the tasks that focus on processing structural data.

3 Our proposed model: DSEDC

In this section, we detail our proposed DSEDC. First, we explain the overall framework of the model, then present each module, and finally, we describe the optimization and complexity analysis.

3.1 Overall framework

The overall framework of our proposed DSEDC model is shown in Fig. 1. It consists of three modules, in particular, the external representation learner (ERL) module, the enhanced internal representation learner (EIRL) module, and the document clustering (DC) module. Given an input document data sample X , the ERL module is designed with the GCN model to learn the external document representation for X . The GCN model is guided by its closeness to generate a reasonable data partition, which is conducted in the DC module. All outputs of the encoder layers of the GCN model are regarded as the external document representation. Specifically, Z is a set of external document representations that consists of all

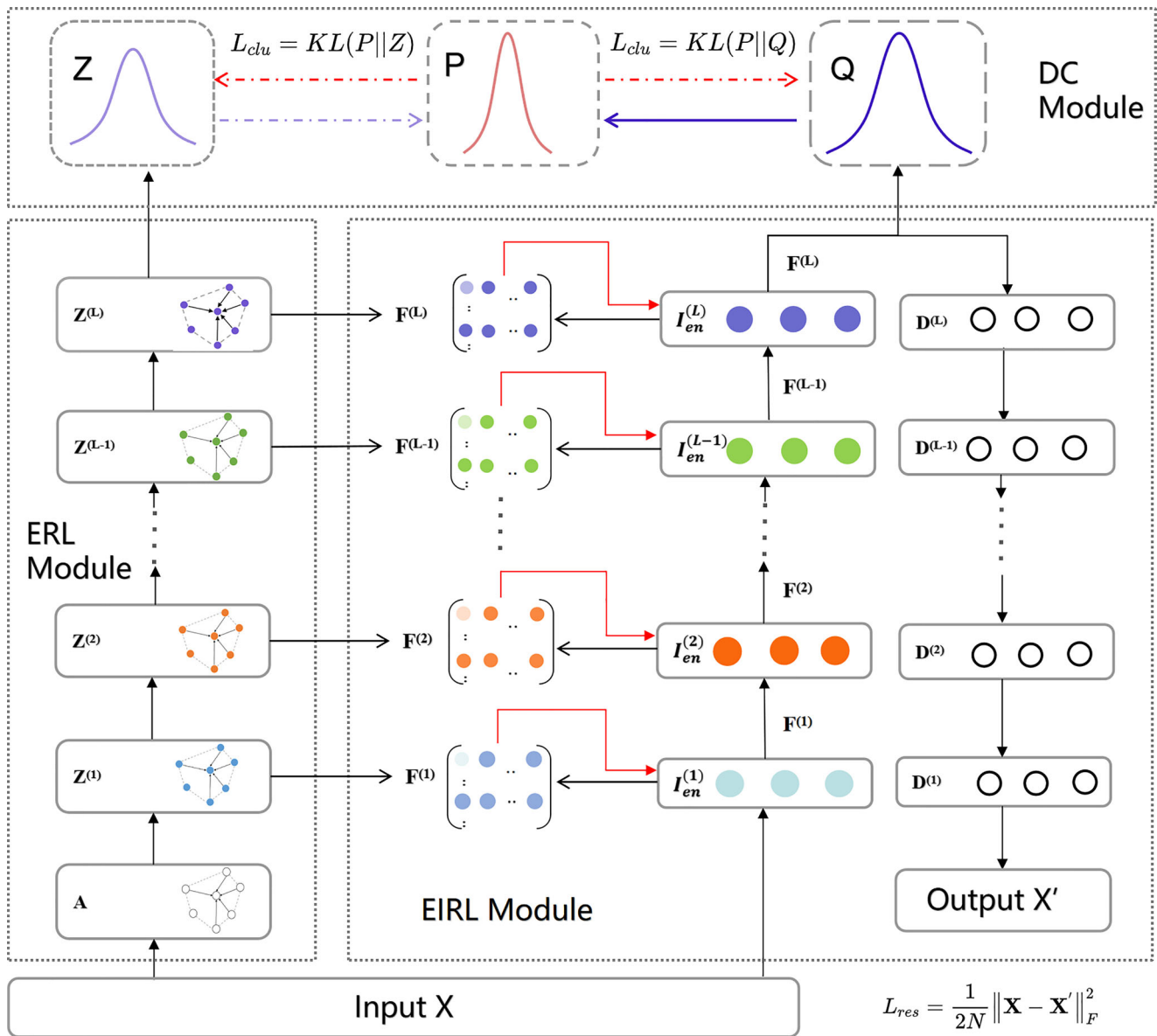


Fig. 1 The DSED model

external semantics learned from different encoder layers in the ERL module. The EIRL module is designed to learn the enhanced internal document representation and the ensemble document representation while considering the interrelated problem of the internal and external representations. Specifically, an AE model is investigated for learning the internal and enhanced internal document representation for each document sample to map the high-dimensional document content features to a low-dimensional feature space. Through the first layer, an ensemble document representation is learned by integrating the internal and external document representations, which contain full semantics of documents collected from both document content and the external environment. The

internal document representation is then improved with the help of the ensemble document representation to form an enhanced internal document representation. The learning of enhanced internal document representation and the ensemble document representation are conducted in a reinforced manner. In particular, we transfer the enhanced internal document representation learned in the l -th layer $I_{en}^{(l)}$ to the corresponding ensemble document representation learning layer $F^{(l)}$. The ensemble document representation learned, $F^{(l)}$, is then used to improve the enhanced internal document representation in the next layer $I_{en}^{(l+1)}$. By transferring the enhanced internal representations layer by layer in the AE model, the final ensemble document representation of the EIRL module is then used as the

input to the DC model to learn the document partition. To further adjust the ERL module, the final output of the external document representation is also imported into the DC model. A joint objective function is designed in the DC model to optimize the document clustering process and fine tune the EIRL and ERL modules.

Notably, in the pretraining process, an identical AE model to the one in the EIRL module is used to initialize the model parameters $\mathbf{W}_e, \mathbf{b}_e, \mathbf{W}_r, \mathbf{b}_r$ for learning the internal semantic representation and the enhanced internal semantic representation by minimizing the reconstruction loss between the raw data and the reconstructed data. Specifically, the pretraining AE model has L layers in the encoder part and corresponding L layers in the decoder part. $\mathbf{W}_e^{(l)}, \mathbf{b}_e^{(l)}$ and $\mathbf{W}_r^{(l)}, \mathbf{b}_r^{(l)}$ are the parameters of the l -layer of the encoder and decoder, respectively.

3.2 The ERL module

In the ERL module, the GCN model is investigated to capture the external structural representation for each document sample. Let l denote the index of the neural network layer. The external document representation $\mathbf{Z}^{(l)}$ captured by the l -th layer of the GCN can be obtained by the following convolutional operation:

$$\mathbf{Z}^{(l)} = f\left(\mathbf{Z}^{(l-1)}, \mathbf{A}|\mathbf{W}_c^{(l-1)}\right), \quad (1)$$

$$f\left(\mathbf{Z}^{(l)}, \mathbf{A}|\mathbf{W}_c^{(l)}\right) = \phi_c\left(\tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-\frac{1}{2}}\mathbf{Z}^{(l)}\mathbf{W}_c^{(l)}\right), \quad (2)$$

$$\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}, \quad (3)$$

$$\tilde{\mathbf{D}}_{ii} = \sum_{j=1}^N \tilde{\mathbf{A}}_{ij}, \quad (4)$$

where $\mathbf{A}$ is the GCN graph structure of each data sample, j is the index of document samples, and N is the number of documents. $\mathbf{I}$ is the identity diagonal matrix of $\mathbf{A}$. $\tilde{\mathbf{D}}_{ii}$ is the degree matrix of $\tilde{\mathbf{A}}$. $\mathbf{W}_c^{(l)}$ is the weight matrix of the l -layer of the ERL module. $f\left(\mathbf{Z}^{(l)}, \mathbf{A}|\mathbf{W}_c^{(l)}\right)$ is a spectral convolution function. ϕ_c is an activation function for the ERL model. Note that the input of the first-layer GCN is the raw document data $\mathbf{X}$.

We employed different strategies to determine the similarity matrix of each data sample for document datasets with or without clear graph structures. For those datasets that with clear graph structures, such as the document datasets with hyperlinks and social network datasets, we directly make use of the graph structure to construct the GCN structure $\mathbf{A}$ for each data sample. For those nongraph document datasets, the KNN model is used to construct the GCN structure graph. In particular, for each document

sample $\mathbf{X}$, K nearest neighbors are selected and their similarity values to $\mathbf{X}$ are used to form $\mathbf{A}$.

3.3 The EIRL module

The ERL module is able to learn the useful external semantic document representation while ignoring the internal representation learned by deep clustering. As mentioned above, the EIRL module is designed to learn an enhanced internal document representation and an ensemble document representation in a reinforced manner. First, an AE model is investigated for learning the internal document representation for each document sample. Second, an ensemble document representation is learned by integrating the internal and external document representations through an ensemble-reinforced enhancement strategy. Finally, with the help of the ensemble document representation, the enhanced internal document representation can be transferred layer by layer in the AE model.

Similar to the pretraining AE model, the neural network for learning the internal document representation is defined as follows.

$$\mathbf{I}^{(l)} = \phi_e\left(\mathbf{W}_e^{(l)}\mathbf{I}^{(l-1)} + \mathbf{b}_e^{(l)}\right), \quad (5)$$

where ϕ_e is the activation function for the encoding layer, and $\mathbf{W}_e, \mathbf{b}_e$ are the model parameters initialized by the pretraining AE model.

With the help of external semantic document representation, the enhanced internal document representation and the ensemble document representation are learned in a reinforced manner as follows:

$$\mathbf{F}^{(l)} = \gamma\mathbf{I}_{en}^{(l)} + (1 - \gamma)\mathbf{Z}^{(l)}, \quad (6)$$

$$\mathbf{I}_{en}^{(l)} = \phi_e\left(\mathbf{W}_e^{(l)}\mathbf{F}^{(l-1)} + \mathbf{b}_e^{(l)}\right), \quad (7)$$

where γ is a balance coefficient factor that adjusts the contribution of internal and external document representations for learning the ensemble document representation $\mathbf{F}$.

Note that the input of the first layer AE is the raw data $\mathbf{X}$, and we denote $\mathbf{I}_{en}^{(1)} = \mathbf{I}^{(1)}$:

$$\mathbf{I}_{en}^{(1)} = \mathbf{I}^{(1)} = \phi_e\left(\mathbf{W}_e^{(1)}\mathbf{X} + \mathbf{b}_e^{(1)}\right). \quad (8)$$

As we can see in (6)(7), through this layered enhancement strategy, the enhanced internal document representation and the ensemble document representation are reinforced layer by layer, so that the semantic information learned by the earlier stage can be used to further enhance the subsequent representation learning process.

The ensemble document representation $\mathbf{F}^{(L)}$ obtained in the last layer of the EIRL module, which contains complete semantics collected internally and externally, is used to learn the document clustering partitions.

We make use of the reconstruction error of each document to fine tune the parameters of the DSEDC model. The reconstruction network is set as follows.

$$\mathbf{D}^{(l)} = \phi_r \left(\mathbf{W}_r^{(l)} \mathbf{D}^{(l+1)} + \mathbf{b}_r^{(l)} \right), \quad (9)$$

where ϕ_r is the activation function for the reconstruction network. $\mathbf{W}_r$ and $\mathbf{b}_r$ are the parameters of the reconstruction network initialized by the pretraining AE model. In addition, we denote $\mathbf{D}^{(L)} = \mathbf{F}^{(L)}$ and $\mathbf{D}^{(0)}$ as the reconstruction of the raw data $\mathbf{X}'$:

$$\mathbf{X}' = \phi_r \left(\mathbf{W}_r^{(1)} \mathbf{D}^{(1)} + \mathbf{b}_r^{(1)} \right). \quad (10)$$

The reconstruction error of the DSEDC model can then be estimated as follows.

$$L_{res} = \frac{1}{2N} \sum_{i=1}^N \|x_i - x'_i\|_2^2 = \frac{1}{2N} \|\mathbf{X} - \mathbf{X}'\|_F^2, \quad (11)$$

where x_i is a raw input document sample, x'_i is a reconstructed document sample, and $\mathbf{X}$ is the raw document data. $\mathbf{X}'$ is the reconstructed data. N is the number of documents.

3.4 DC module

There are two tasks in the DC module. The first task is to learn the document partition with the help of the ensemble document representation $\mathbf{F}^{(L)}$ learned by the EIRL module. The second task is to optimize the document representation learning networks of our proposed DSEDC model. We designed a joint objective function to fulfill the above two tasks simultaneously. The objective function makes use of two types of guide information. Specifically, the confidence of the clustering partition and the reconstruction error of the document data sample. However, the ERL module and EIRL module cannot be applied to the document clustering tasks directly. Therefore, we design a joint objective function to learn the clustering assignment and guide the update of the whole model. For learning the document partition, given the i -th document sample and the j -th cluster, we use the Student's-distribution [27] as a kernel to measure the similarity between the ensemble document representation f_i and the cluster center μ_j , and we treat $Q = [q_{ij}]$ as the clustering result distribution. In particular, to improve the cluster cohesion, we calculate the clustering result distribution Q and the target distribution P as follows:

$$q_{ij} = \frac{\left(1 + \|f_i - \mu_j\|^2 / \alpha\right)^{-\frac{\alpha+1}{2}}}{\sum_{j'=1}^J \left(1 + \|f_i - \mu_{j'}\|^2 / \alpha\right)^{-\frac{\alpha+1}{2}}}, \quad (12)$$

$$p_{ij} = \frac{q_{ij}^2 / \tau_j}{\sum_{j'=1}^J q_{ij'}^2 / \tau_{j'}}, \quad (13)$$

where $\tau_j = \sum_{i=1}^N q_{ij}$ is the soft cluster frequency. j is the index of the cluster, and J is the number of clusters. α is the degree of freedom of Student's-distribution. To make the distribution with higher confidence, we can derive our objective function for document clustering as follows:

$$L_{clu} = KL(P \| Q) = \sum_{i=1}^N \sum_{j=1}^J p_{ij} \log \frac{p_{ij}}{q_{ij}}. \quad (14)$$

To further optimize the parameters of the ERL module, we also use the same method to measure the similarity between the external representation z_i and the cluster center μ_j . The distribution Z is calculated as follows:

$$z_{ij} = \frac{\left(1 + \|z_i - \mu_j\|^2 / \alpha\right)^{-\frac{\alpha+1}{2}}}{\sum_{j'=1}^J \left(1 + \|z_i - \mu_{j'}\|^2 / \alpha\right)^{-\frac{\alpha+1}{2}}}, \quad (15)$$

where Z indicates the clustering result distribution produced by $\mathbf{Z}^{(L)}$. For training the ERL module, similar to [25], we use the target distribution P to guide distribution Z . The objective function of tuning the GCN model is designed as follows.

$$L_{gcn} = KL(P \| Z) = \sum_{i=1}^N \sum_{j=1}^J p_{ij} \log \frac{p_{ij}}{z_{ij}}. \quad (16)$$

Through (16), the target distribution P connects the distribution Z and Q . Therefore, the enhanced internal semantics are used to improve the external document representations at this stage.

With the two above objective functions and the reconstruction loss, we design the joint objective function for the DSEDC model as follows.

$$L = L_{clu} + w_r L_{res} + w_g L_{gcn}, \quad (17)$$

where w_r and w_g are parameters to balance the clustering optimization. $w_r \in [0, 1]$ controls the weight of the reconstruction objective of the DSEDC. $w_g \in [0, 1]$ influences the disturbance of the GCN.

3.5 Optimization

In the optimization stage, our model uses the ranger optimizer [28], which fuses the RAdam optimizer [29] and the LookAhead optimizer [30] together, to optimize both the clustering center and the network parameters. This optimizer maintains a set of slow weights φ and fast weights ϵ , which are synced with the fast weights every s updates.

After the s inner optimizer updates using optimizer O , the slow weights are updated toward the fast weights by linearly interpolating in weight space, $\epsilon - \varphi$.

The gradients of the objective function L_{clu} of the ensemble document representation f_i and μ_j are calculated as follows:

$$\frac{\partial L_{clu}}{\partial f_i} = \frac{\alpha + 1}{\alpha} \sum_{j=1}^J (1 + \|f_i - \mu_j\|)^{-1} \times (p_{ij} - q_{ij}) (f_i - \mu_j), \quad (18)$$

$$\frac{\partial L_{clu}}{\partial \mu_j} = \frac{\alpha + 1}{\alpha} \sum_{i=1}^J (1 + \|f_i - \mu_j\|)^{-1} \times (q_{ij} - p_{ij}) (f_i - \mu_j). \quad (19)$$

The gradient of the objective function L_{gcn} of the external semantic representation z_i is calculated as follows:

$$\frac{\partial L_{gcn}}{\partial z_i} = \frac{\alpha + 1}{\alpha} \sum_{j=1}^J (1 + \|z_i - \mu_j\|)^{-1} \times (p_{ij} - z_{ij}) (z_i - \mu_j). \quad (20)$$

3.5.1 Update the model's fast weights and cluster centers

We set the RAdam optimizer as optimizer O then, the fast weights and cluster centers of our model can be calculated as follows:

$$\epsilon_{t,i+1} = \epsilon_{t,i} + O(L, \epsilon_{t,i-1}, \beta), \quad (21)$$

where β represents the current mini-batch, $\epsilon_{t,i}$ contains all the parameters of our model and z_i, f_i, μ_j .

3.5.2 Update the model's slow weights and cluster centers

The slow weights and cluster centers of our model can be calculated as follows:

$$\varphi_{t+1} = \varphi_t + \nu (\epsilon_{t,s} - \varphi_t), \quad (22)$$

where ν represents the slow weight learning rate. After each slow weight update, the fast weights are reset to the current slow weight value.

3.5.3 Update target distribution

In practice, because the representation learned by the EIRL is the enhanced representation in our model, we choose the

distribution Q as our final clustering result. Then the label estimated for data sample x_i can be obtained as follows:

$$r_i = \arg \max_j q_{ij}, \quad (23)$$

where q_{ij} is calculated in (12). Algorithm 1 shows the training process of the whole model.

Input: document data $\mathbf{X}$, number of neighbors K , number of clusters J , Maximum iterations M

Output: clustering results R

- 1: Constructing initial structure-graph $\mathbf{A}$ by data $\mathbf{X}$;
 - 2: Initialize $\mathbf{W}_e^{(l)}, \mathbf{b}_e^{(l)}, \mathbf{W}_r^{(l)}, \mathbf{b}_r^{(l)}$ with pre-train AE;
 - 3: Initialize $\mathbf{W}_c^{(l)}$ randomly;
 - 4: Initialize μ with K-means on the representation learned by pre-train AE;
 - 5: **for** $iter \in 0, 1, \dots, M$ **do**
 - 6: Generate GCN representations $\mathbf{Z}^{(l)}, l \in [1, L]$;
 - 7: Use $\mathbf{Z}^{(L)}$ to compute distribution Z via (15);
 - 8: **for** $l \in 1, 2, \dots, L$ **do**
 - 9: Use the representation enhancing operator:
 - 10: $\mathbf{F}^{(l)} = \gamma \mathbf{I}_{en}^{(l)} + (1 - \gamma) \mathbf{Z}^{(l)}$;
 - 11: Generate the next layer enhanced internal representation :
 - 12: $\mathbf{I}_{en}^{(l)} = \phi_e(\mathbf{W}_e^{(l)} \mathbf{F}^{(l-1)} + \mathbf{b}_e^{(l)})$;
 - 13: **end**
 - 14: Use $\mathbf{F}^{(L)}$ to compute distribution Q via (12);
 - 15: Calculate target distribution P via (13);
 - 16: Feed $\mathbf{D}^{(L)}$ to the decoder to construct the raw data $\mathbf{X}$;
 - 17: Calculate $L_{clu}, L_{res}, L_{gcn}$, respectively;
 - 18: Calculate the loss function via (17);
 - 19: Back propagation and update parameters in DSEDC;
 - 20: **end**
 - 21: Calculate the clustering results based on distribution Q ;
 - 22: Return R ;
-

Algorithm 1 Training process of DSEDC.

3.6 Complexity analysis

In this work, we will analyze the complexity of the proposed model. We set d as the dimension of the input data $\mathbf{X}$, denote the data dimension of each layer in the AE as $d_1, d_2, \dots, d_L$, and N as the number of input data $\mathbf{X}$. For the AE, the time complexity is $\mathcal{O}(Nd^2d_1^2d_2^2 \dots d_L^2)$. For the GCN, the time complexity is linearly related to the number of edges $|\mathcal{E}|$. Therefore, the time complexity of the GCN is $\mathcal{O}(|\mathcal{E}|dd_1d_2 \dots d_L)$. For the objective function, we assume J as the number of clustering categories. According to the result in [1], the time complexity of

(12) is $\mathcal{O}(NJ + N \log N)$. The complexity of Algorithm 1 is $\mathcal{O}(Nd^2d_1^2d_2^2 \dots d_L^2 + |\varepsilon|dd_1d_2 \dots d_L + NJ + N \log N)$, which is linearly related to the numbers of samples and edges.

4 Experiments

4.1 Experimental datasets

We conduct our experiments on five real-world document datasets. These datasets can be categorized into two groups: documents with links and without links. The statistics of these datasets are shown in Table 1. The detailed descriptions are reported as follows.

- ACM¹ [31]: It is a document dataset with links from the ACM website. Documents features are the bag-of-words of the keywords and a list of coauthor links between documents. Papers published in KDD, SIGCOMM, SIGCOD, and MobiCOMM are selected and divided into three research classes(database, wireless communication, data mining) in this paper.
- Citeseer²: It is a document dataset with links from the Citeseer website. Each document contains sparse bag-of-words feature vectors and a list of citation links between documents. The labels contain six areas: agents, artificial intelligence, database, information retrieval, machine language, and HCI.
- Reuters [25]: It is a web text document dataset containing approximately 810000 English news stories labeled with a category tree. We use four root categories, corporate/industrial, government/social, markets and economics, as labels and sample a random subset of 10000 examples for clustering.
- BBC³: BBC is a web text document dataset containing five classes of news: business, entertainment, politics, sport, and tech. It consists of 2225 documents from the BBC news website corresponding to stories in five topical areas from 2004-2005.
- Aminer-s⁴: This is a text document dataset. We downloaded research papers published on the Aminer website and used the abstract of each paper. All features of these documents are preprocessed by the tf-idf scheme. It contains three research areas: infocoms, database, and graphics.

Table 1 The statistics of the document datasets with links

Dataset	Samples	links	Classes	features
ACM	3025	13128	3	1870
Citeseer	3327	4732	6	3703
Reuters	10000	–	4	2000
BBC	2225	–	5	9635
Aminer-s	4306	–	3	10000

4.2 Baseline methods

These baseline methods can be categorized into three groups: methods using internal representation only (K-means [32], AE [5], VAE [33], DEC [1], IDEC [2]), methods using external representation only (Spectral [16], DNGR [17]), methods using both internal representation and external representation (GAE [19], VGAE [19], ARGVGA [20], ARVGA [20], SDCN and SDCN_Q [25], AGCN [26]). The detailed descriptions are reported as follows.

- Methods using internal representation only:
 - K-means [32] is a traditional clustering model without deep learning.
 - AE [5] and VAE [33] are deep clustering models that learn representations by an autoencoder and variational autoencoder, respectively.
 - DEC [1] is a deep clustering model that designs a clustering objective to guide the learning of the data representations.
 - IDEC [2] adds a reconstruction loss to DEC to learn better representations.
- Methods using external representation only:
 - Spectral [16] treats the samples as the nodes in a weighted graph and uses the graph structure of data for clustering.
 - DNGR [17] is a spectral clustering method that uses the adjacency matrix as the similarity matrix.
- Methods using both internal representation and external representation:
 - GAE and VGAE [19] are unsupervised graph embedding models using GCN to learn representations on the autoencoder and variational autoencoder respectively.
 - ARGVGA and ARVGA [20] are variants of GAE and VGAE by adding an adversarial model respectively.
 - SDCN [25] is a structural deep clustering model that uses a GCN and DNN to jointly construct a clustering network.
 - SDCN_Q [25] is the variant of SDCN with distribution Q .
 - AGCN [26] is an attention-driven graph clustering model that dynamically fuses the internal and external

¹<http://dl.acm.org/>

²<http://citeseerx.ist.psu.edu/index/>

³<http://mlg.ucd.ie/datasets/bbc.html>

⁴<https://www.aminer.cn>

representations and adaptively aggregates the multi-scale features embedded at different layers.

We investigated a set of DSEDC-related models to evaluate the effectiveness of the layered enhancement strategy. For each DSEDC-related model, we select different layers to transfer the external document representation from the ERL module to the EIRL module for representation enhancement. There are two different directions to conduct the representation enhancing, in particular, enhance from the beginning layer, denoted as “DSEDC-S”, and enhance from the last layer, denoted as “DSEDC-E”. Note that if there are no layers selected to transfer the external document representations to the EIRL module, the ERL and EIRL modules become two separated processes. If all layers are selected, the ERL and EIRL modules of the DSEDC model are fully connected. The layered enhancement strategy is completely conducted. In our experiments, we set the total number of layers of the ERL and EIRL modules to 4. We investigated 6 variants of the DSEDC model, in particular the DSEDC-S1, DSEDC-S2, DSEDC-S3, DSEDC-E1, DSEDC-E2, and DSEDC-E3 models. DSEDC-S l denotes the DSEDC-related model that enhances l layers of ERL and EIRL modules from the beginning layer. DSEDC-E l denotes the DSEDC-related model that enhances l layers of ERL and EIRL modules from the last layer.

4.3 Evaluation metrics and experimental setup

To measure the performance of the document clustering task, we utilize three popular clustering evaluation metrics [25, 34]: accuracy(ACC), normalized mutual information(NMI), and average Rand index(ARI). For each evaluation metric, a higher value implies better performance. Denote $Y = \{y_1, y_2, \dots, y_N\}$ as the ground-truth label and $C = \{c_1, c_2, \dots, c_N\}$ as the predicted cluster label. These 3 metrics can be calculated as follows:

- **Accuracy (ACC)** is formulated as follows:

$$ACC = \max_m \frac{\sum_{i=1}^N \{y_i = m(c_i)\}}{N}, \quad (24)$$

where $m(\cdot)$ is a mapping function that ranges over all possible one-to-one mappings between the true label Y and predicted cluster label C .

- **Normalized mutual information (NMI)** is used to measure the normalized similarity of the true label Y and the corresponding predicted cluster label C and can be represented as follows:

$$NMI = \frac{I(Y, C)}{\max[H(Y), H(C)]}, \quad (25)$$

where I is the mutual information between the true label Y and predicted cluster label C . $H(Y)$ and $H(C)$

are the entropy of the true label Y and predicted cluster label C .

- **The average Rand index (ARI)** is an effective method for comparing two partitions and can be formulated as follows:

$$ARI = \frac{RI - E[RI]}{\max(RI) - E[RI]}, \quad (26)$$

where $RI = \frac{A+B}{U_2^N}$. $E[RI]$ is the expected value of RI when the partitions are made at random. U_2^N presents all the possible combination pairs of samples. A represents the number of sample combination pairs in the same partition both Y and C , and B represents the number of sample combination pairs not in both Y and C .

For the methods(AE, DEC, IDEC, SDCN, AGCN, and DSEDC) that have a pretraining process, we use the same dimension d -500-500-2000-10, where d is the dimension of the input data. We train the autoencoder using all data samples with 30 epochs and the learning rate 10^{-3} by the Adam algorithm. For the methods and datasets used in the baseline methods, we use the results reported in corresponding papers. For other baseline methods, we leave the settings of as default mentioned in corresponding papers. For our model, we set the dimensions of GCN and AE to d -500-500-2000-10, $K = 30$ on all the nongraph datasets, and $\alpha = 2$, $\gamma = 0.5$, $w_g = 0.01$ and $w_r = 0.1$ for all experiments because our model is not sensitive to hyperparameters when it is used in different datasets. We train the DSEDC for 3,000 epochs and optimize them with the learning rate 10^{-4} by the Ranger algorithm. We repeat all the methods 10 times and then report the average results to prevent extreme cases.

4.4 Clustering results

This section provides the results of our model on multiple datasets with comparisons to the baselines. Table 2 shows the clustering results on real-world document datasets with and without links. Our observations are as follows:

For all metrics, DSEDC and its variants show superior performance to baseline methods by a considerable margin on both the documents with links and without links. Such results show strong evidence advocating our proposed DSEDC model. The improvements on the Reuters and BBC datasets are more substantial, competing with the strongest baseline method, our model outperforms it by 3.71%, 15.43% and 12.03% on BBC, 9.05%, 9.23% and 13.76% on Reuters with respect to ACC, NMI and ARI. The reason is that the quality of the original document representation is poor in these datasets, leading to poor document clustering performance. The improvement on the Aminer-s dataset is smaller than those of the others because the quality of the

Table 2 Clustering results on real-world document datasets

Methods	Reuters			BBC			ACM			Citeseer			Aminer-s		
	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI
K-means	54.04	41.54	27.95	51.15	32.97	14.37	67.31	32.44	30.60	39.32	16.94	13.43	69.18	38.26	27.69
AE	74.90	49.69	49.55	51.73	51.33	37.33	81.83	49.30	54.64	57.08	27.64	29.30	75.56	45.26	39.95
VAE	61.33	33.20	19.79	70.23	53.04	43.04	69.75	31.16	32.53	39.26	15.47	14.88	81.17	50.64	51.15
DEC	73.58	47.50	48.44	77.06	61.51	59.59	84.33	54.54	60.64	55.89	28.34	28.12	85.25	57.15	61.02
IDEC	75.43	50.28	51.26	76.40	64.48	61.72	85.12	56.61	62.16	60.49	27.17	25.70	92.65	72.63	78.98
Spectral	56.23	33.68	22.09	71.96	61.68	43.73	73.62	40.63	40.89	46.98	22.85	21.44	43.01	14.91	2.84
DNGR	54.95	29.29	25.20	76.90	56.21	50.16	37.69	1.76	1.63	32.59	18.02	4.29	54.16	14.59	11.33
GAE	54.40	25.92	19.61	79.75	65.54	61.69	84.52	55.38	59.46	61.35	34.63	33.55	87.37	59.00	65.32
VGAE	60.85	25.51	26.18	79.91	63.43	63.23	84.13	53.20	57.72	60.97	32.69	33.13	88.27	61.26	67.49
ARGA	48.87	15.13	14.19	84.31	61.56	65.16	83.50	51.09	56.04	57.30	35.00	34.10	87.39	58.19	65.49
ARVGA	54.53	25.39	13.28	85.44	64.73	66.75	85.45	54.86	61.22	54.40	26.10	24.50	85.56	56.90	60.35
SDCN _Q	79.30	56.89	59.58	77.02	62.97	60.91	86.95	58.90	65.25	61.67	34.39	35.50	91.82	71.35	76.79
SDCN	77.15	50.82	55.36	77.55	65.28	62.58	90.45	68.31	73.91	65.96	38.71	40.17	93.62	75.62	81.65
AGCN	79.30	57.83	60.55	72.60	62.98	57.38	90.59	68.38	74.20	68.79	41.54	43.79	93.33	74.65	80.80
DSEDC-E1	81.29	60.67	62.58	66.85	52.76	46.82	90.19	67.39	73.05	59.21	32.32	32.48	94.50	78.12	84.06
DSEDC-E2	81.60	61.37	63.48	76.07	65.06	62.46	92.06	73.14	77.99	60.71	33.55	34.27	94.74	78.93	84.73
DSEDC-E3	86.23	63.52	69.34	79.84	68.30	65.94	92.17	73.46	78.29	62.83	36.52	37.41	95.53	81.33	86.91
DSEDC-S1	81.16	60.00	62.17	66.84	53.89	47.86	89.68	65.88	71.77	59.68	32.58	33.01	94.87	79.48	85.08
DSEDC-S2	81.42	60.52	62.87	81.39	69.62	66.98	92.26	73.41	78.50	61.64	35.11	35.80	95.52	81.53	86.87
DSEDC-S3	86.77	63.58	69.51	86.16	72.25	71.25	92.32	73.72	78.66	63.52	37.28	38.34	95.62	81.77	87.18
DSEDC	86.48	63.17	68.88	88.61	75.35	74.78	91.33	70.99	76.14	70.37	44.16	45.37	95.78	82.29	87.60

The bold numbers represent the best results

Aminer-s dataset is better. Most baseline methods achieve relatively good results, which implies that the quality of those features in the original representation is useful for discovering document structures. This approach leaves little space for the semantic representation to improve. However, our proposed DSEDC model still performs the best on the Aminer-s dataset.

Comparing the results of the baselines, it is obvious that methods using both internal and external semantic representations usually achieve better performance than methods leveraging information from a single source. In particular, the DNGR performs worse on a few datasets. The graph construction of the DNGR relies on the relative distance of node features to control the context weight, which may be inconsistent with the graph constructed by external information (such as coauthor in ACM, citation in Citeseer).

The clustering results of the DSEDC variants prove that the DSEDC benefits from the layered enhancement strategy. Increasing the number of DSEDC layers substantially enhances the clustering performance. Obviously, on all datasets, DSEDC-S2, DSEDC-S3, DSEDC-E2, DSEDC-E3, and DSEDC have achieved consistent improvements

over DSEDC-S1 and DSEDC-E1, which only use one layer to integrate the external semantic representation. In addition, DSEDC performs better than other methods in three datasets (BBC, Citeseer, Aminer-s). As the number of enhanced layers increases, the clustering performance gradually increases. This implies that for these datasets, more meaningful external semantic representations are obtained layer by layer through our layered enhancement strategy. There is an interesting phenomenon in which the performance of DSEDC is not as good as DSEDC-S3 and DSEDC-E3 in Reuters dataset and not as good as those of DSEDC-S3, DSEDC-E3, DSEDC-S2 and DSEDC-E2 in the ACM dataset. For the Reuters dataset, compared with DSEDC-E3, DSEDC has the additional external semantic representation $Z^{(1)}$ of the first layer. Compared with DSEDC-S3, DSEDC has the additional external semantic representation of $Z^{(4)}$ in the last layer. Due to the large scale of samples or links, the $Z^{(1)}$ that directly comes from the original graph and the $Z^{(4)}$ without the learning process may introduce noise. For the ACM dataset, as before, because of its large number of links between documents, discovering the additional external representation may also introduce noise at the same time.

4.5 Ablation study

To validate that each module of the DSEDC has its own contribution, we conduct an ablation study on the Citeseer dataset to manifest the efficacy of the internal, external, and ensemble document representations and the joint objective function in the DSEDC. We set four variants of the DSEDC for comparison. KL means the method only has been supervised by the traditional loss function $L = L_{clu} + w_r L_{res}$. “IR+KL” and “ER+KL” cluster raw documents using the internal representation and the external representation, respectively. “IR+ER+RF+KL” utilizes the ensemble document representations for document clustering, and “IR+ER+RF+DC” further adds the joint objective function and is exactly the full model. In the Table 3, it is obvious that each part of our model contributes to the final performance, which evidently states their effectiveness of them. Additionally, the model “IR+ER+RF+KL” outperforms all the baselines, verifying to the layered enhancement strategy of our internal and external semantic representation learning.

4.6 Visualization

4.6.1 Visualization of the different semantic representations and clustering performance

To intuitively show the layered enhancement strategy can significantly improve the internal and external semantic representations during the enhancing process layer by layer, and well cluster the documents according to their corresponding classes. We visualize the representations in 2D space using the *t-SNE* method [35] on the BBC dataset. The figures are shown in Fig. 2. Lines 1-3 depict the internal, external and ensemble document representations of different layers learned in DSEDC, respectively. The last line depicts the final clustering performance. As the number of document representations learned layer increases, the cluster property of the representation becomes increasingly obvious. Compared with the ensemble document representation learned by each layer, the internal and external

representations are obviously loose, and there are many dots sticking to the edge of other clusters. The ensemble document representation is denser. It is obvious that documents are more clearly separable by using the DSEDC model.

4.6.2 Visualization of enhancing semantics by different enhanced layers

For a more intuitive comparison, we visualize the clustering result in Fig. 3. We find that increasing the number of enhanced layers substantially enhances the semantic learning of documents. From Fig. 3, it is obvious that the documents clusters of DSEDC-S0 are loose with unclear cluster boundaries. As the number of layers increases, the cluster property becomes increasingly obvious. The cluster information is clearly displayed. Clusters of the DSEDC model are denser, and documents are more clearly separable.

4.7 Analysis of the parameters and training process

4.7.1 Analysis of the number of neighbors K

The number of neighbors K is a significant parameter in constructing the KNN graph for capturing the external information. We explore the clustering performance of DSEDC by varying the values of K , and $K \in \{1, 5, 10, 30, 50\}$ on nongraph document datasets (BBC, Reuters, Aminer-s). From Fig. 4, it is obviously that the document clustering performance shows rapid improvement with the addition of a few K . K gradually reached the optimal value. When K increases to a relatively large value, the results of document clustering are flat with a slow descending tendency due to the introduction of noise information with more neighbors on different datasets. Therefore, we set the value of K for all experiments within a reasonable range.

4.7.2 Analysis of balance coefficient γ

To study the effect of balance coefficient γ , we set $\gamma = \{0.0, 0.1, 0.3, 0.5, 0.7, 0.9, 1.0\}$ on all datasets in detail. Note that $\gamma=0$ means that the learned semantic representation only contains the external representation, and $\gamma=1$ represents that the DSEDC only uses the internal semantic representation.

From Fig. 5, all metrics with parameter $\gamma=0.5$ achieve the best performance in four datasets (BBC, Reuters, Citeseer, Aminer-s), which shows that the internal and the external semantic representation are equally important and that the improvement of DSEDC depends on the ensemble document representation. Another interesting observation is that when $\gamma = 0.1$ and $\gamma = 0.9$, we find that even fusing a small amount of internal or external representations

Table 3 Ablation study

Model Variants	Input	Citeseer		
		ACC	NMI	ARI
IR+KL	IR	58.97	31.18	31.69
ER+KL	ER	36.50	18.52	14.21
IR+ER+RF+KL	IR+ER	68.65	42.19	43.74
IR+ER+RF+DC	IR+ER	70.37	44.16	45.37

The bold numbers represent the best results

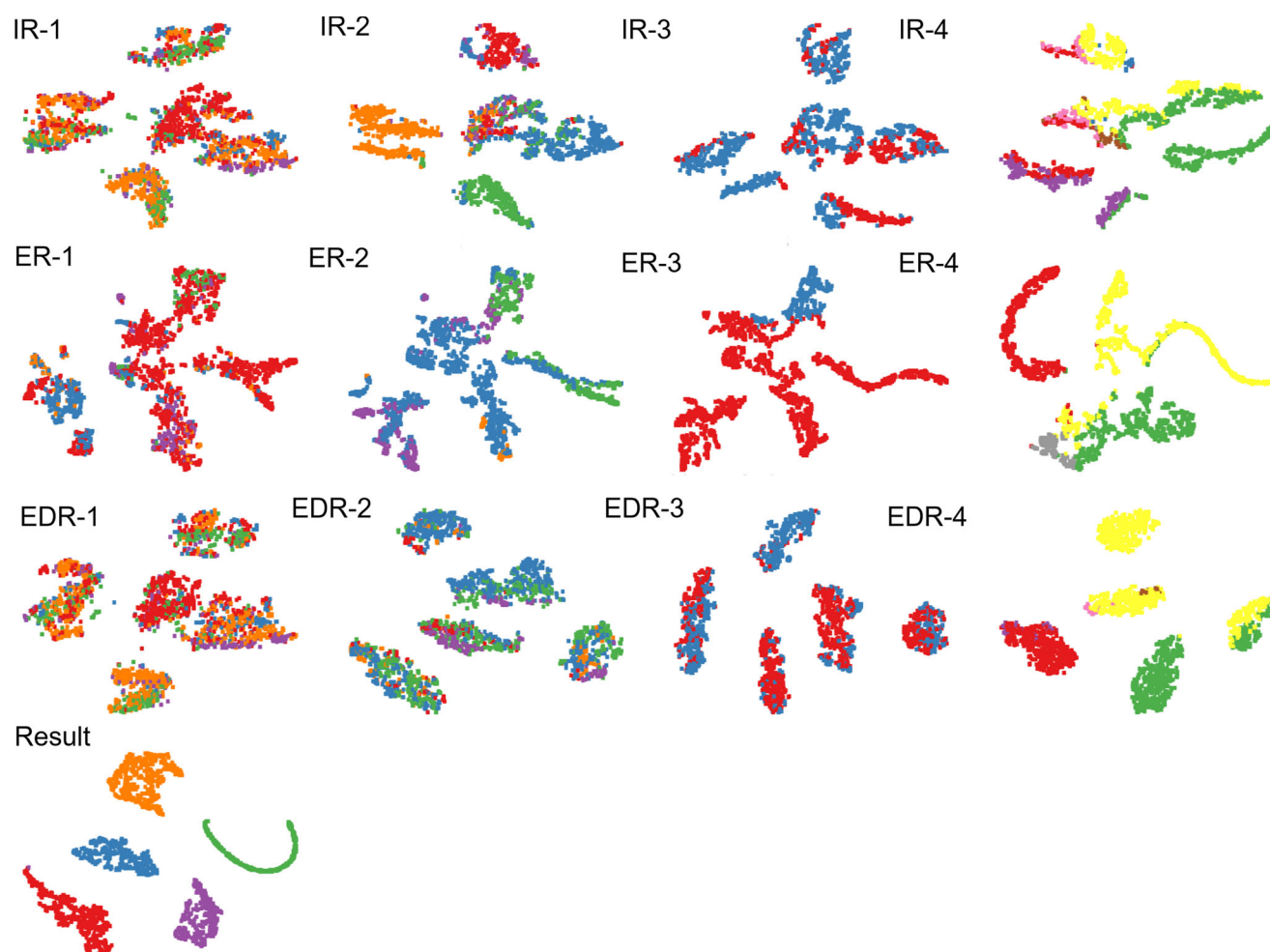


Fig. 2 Visualization of the different semantic representations

can improve the document clustering performance, and also help alleviate the oversmoothing problem.

4.7.3 Analysis of the training process

To explore how the cluster performance varies with the number of iterations, we analyze the training progress

on all datasets in this section. In Fig. 6, the results of all metrics begin with poor performance. Because the representations learned by the EIRL and ERL modules are different, a conflict may arise between the results of the two modules, resulting in low clustering results. Then, all metrics gradually increase to a high level because the DC module eases the conflict between the two modules, making

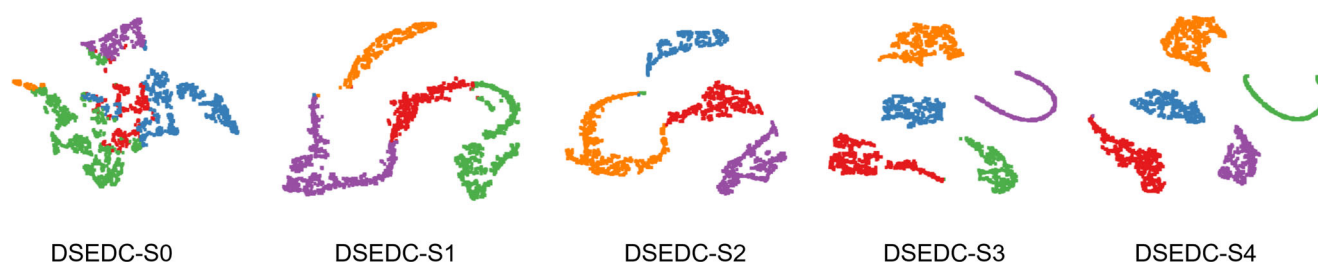


Fig. 3 Visualization of clustering results with different enhanced layers

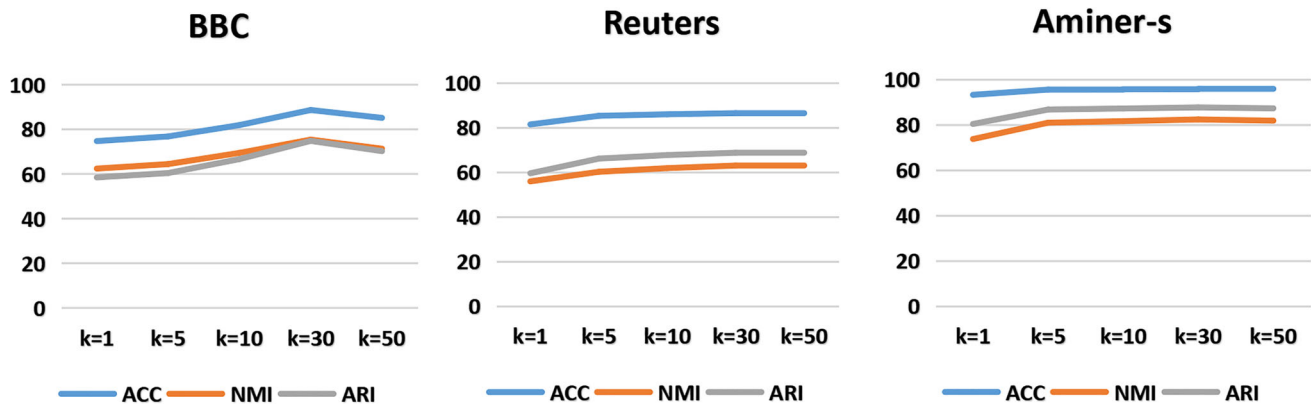


Fig. 4 Effect of different number of neighbors K



Fig. 5 Effect of the balance coefficient γ

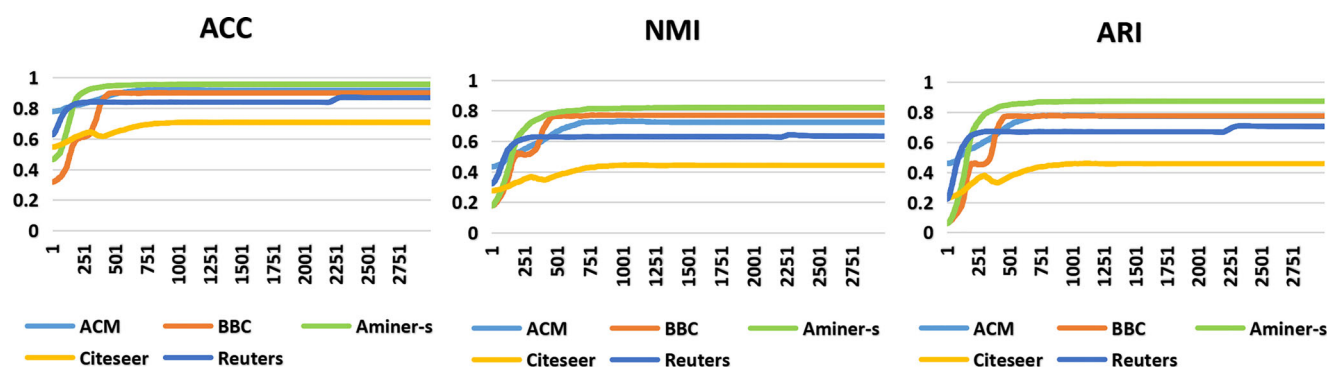


Fig. 6 Analysis of Training Process

the clustering results tend to be consistent. It is obvious that with the increase in training epochs, the clustering results of DSEDC tend to be stable, and there is no significant fluctuation, indicating the good robustness of our proposed model.

5 Conclusion

In this paper, a novel end-to-end deep document clustering model, namely, DSEDC, was presented to solve the problem of document data semantic representation enhancement with the help of a structural enhanced network. An ensemble-reinforced enhancement strategy was developed to learn enhanced internal document representation as well as ensemble document representation simultaneously. A joint objective function was designed to simultaneously optimize the document representations and the data clustering. Experimental results on various document datasets show that our DSEDC is effective.

An interesting direction for future research is to study how to use the DSEDC model to handle the noise features of text documents, which is also a crucial problem of document analysis. The idea of adding complementary useful semantic information can be explored to minimize those confusing noise features.

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